

Information Theory for Probabilistic Modeling

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1 Outline

The aim of this document is to formalize and substantiate these claims:

1. entropy is the compressibility of a discrete iid source, and entropy rate is the compressibility of a discrete stationary process (Section 2 here, Chapters 2,4, and 5 in C&T);
2. predictive likelihood on data is a model's ability to compress that data (Section 3);
3. continuous (differential) entropy loses some direct interpretations, though it allows relative comparisons (Section 4 here, Chapter 8 in C&T); and
4. any probabilistic model that can predict can be made into a code (compression scheme), though you can skip the model-fitting step if you don't want to make predictions or interpret the model (Section 5 here, Chapters 5 and 13 in C&T).

Wherever possible, there are citations to sections, theorems, and examples from Cover et al. (2006), referenced as C&T, which is the standard text on information theory. Proofs are shown here only when they're actually enlightening.

2 Entropy

The entropy of a discrete random variable is a non-negative number that depends on the random variable's distribution. It is zero iff the random variable is deterministic, and it is larger when the random variable is more unpredictable.

Definition 1 (Entropy, C&T Section 2.1). *For a random variable X taking values in $\mathcal{X}$ with pmf p , the entropy of X is*

$$H(X) := - \sum_{x \in \mathcal{X}} p(x) \log p(x) = \mathbb{E}_{x \sim p}[-\log p(x)] \quad (1)$$

with the convention that $0 \log 0 = 0$ and that $\log = \log_2$ unless otherwise specified. Sometimes we write $H(p)$ for $H(X)$ when $X \sim p$.

See C&T Example 2.1.1 and Figure 2.1. Chapter 2 is all about entropy.

When the log in Eq. (1) is base 2, the units are *bits* as motivated by the connection to compression described next.

When a random variable takes values in a discrete set $\mathcal{X}$, one can encode samples of the random variable using $\lceil \log |\mathcal{X}| \rceil$ bits per sample, but when a random variable is predictable it may be possible to use fewer bits: one could assign short descriptions to frequent symbols and longer descriptions for infrequent symbols, like in Morse Code. The next two claims show that entropy of a random variable essentially equals the average number of bits required to encode its samples.

First, for any encoding¹ of the symbols of $\mathcal{X}$ as bit strings, the entropy $H(X)$ lower-bounds the average bit string length.

¹This argument only applies to prefix codes, but by the Asymptotic Equipartition Property (C&T Chapter 3) the story is the same for all codes.

Claim 1 (Entropy lower-bounds average codeword length, Thm 5.3.1). *If ℓ_i is the bit string length for encoding x_i and $L(x_i) = \ell_i$, then*

$$H(X) \leq \mathbb{E}[L(X)].$$

Proof. The key idea is to think of symbols' bit string lengths as corresponding to a probability distribution for which they are optimal codes; that is, think of a new probability mass function q with

$$q_i \propto 2^{-\ell_i}, \text{ i.e. } q(x_i) := \frac{e^{-\ell_i}}{C} \text{ where } C := \sum_j 2^{-\ell_j}$$

so that $\ell_i = -\log q_i + \log C$. Then comparing a code's average length can be rewritten as comparing two probability distributions by substituting for ℓ_i :

$$\mathbb{E}[L(X)] - H(X) = \sum_i p_i \ell_i + \sum_i p_i \log p_i \quad (2)$$

$$= \sum_i p_i \log p_i - \sum_i p_i \log q_i + \log C \quad (3)$$

$$= \underbrace{\sum_i p_i \log \frac{p_i}{q_i}}_{D(p||q)} + \log C \quad (4)$$

$$\geq 0$$

Where $D(p||q) \geq 0$ is known as Gibbs' inequality (proved by $x - 1 \geq \ln x$) and $\log C \geq 0$ is from the Kraft inequality C&T Section 5.2 (proved by tree representation of prefix codes). $\square$

Note that $D(p||q)$ is a pseudometric between distributions, so that it is zero iff $p = q$ almost everywhere and positive otherwise; therefore Eq. (4) hints that choosing a code for a distribution q instead of the true distribution p incurs a compression cost of $D(p||q)$ bits per symbol. For optimal codes, C asymptotically goes to 1 so the $\log C$ term goes to zero. (See C&T Thm 5.4.3.)

The previous claim shows that entropy tells us the best bits per symbol we can hope for; the next claim shows that it's not too hard to get within one bit per symbol of that limit.

Claim 2 (Average coded symbol length can be within one bit of entropy, Theorem 5.4.1). *It's not hard to choose L so that*

$$\mathbb{E}[L(X)] \leq H(X) + 1$$

Proof. By choosing² $\ell_i = \lceil \log \frac{1}{p_i} \rceil$ we get an average length of

$$\begin{aligned} \mathbb{E}[L(X)] &= \sum_i p_i \lceil \log \frac{1}{p_i} \rceil \\ &\leq \sum_i p_i \left(-\log \frac{1}{p_i} + 1 \right) \\ &= -\sum_i p_i \log p_i + 1 \\ &= H(X) + 1 \end{aligned}$$

and an optimal code can only do better (C&T Section 5.4). $\square$

²Always possible by reversing the argument in the Kraft inequality proof C&T Section 5.2

The discussion so far has been about compressing a stream of independent samples from some probability distribution, but the same story holds for streams where the symbols are dependent. In particular, for streams that are stationary stochastic processes (like Markov chains or, pretty much generally, hidden Markov models). The key is to measure then *entropy rate* of the process, which reduces to entropy in the independent case.

Definition 2 (Entropy rate, C&T Section 4.2). *The entropy rate of a stochastic process $X = (X_i)$ is*

$$\begin{aligned} H(X) &:= \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1, X_2, \dots, X_n) \\ &= \lim_{n \rightarrow \infty} H(X_n | X_{n-1}, \dots, X_1) \end{aligned}$$

where the second equality only holds for stationary processes.

For a Markov chain with invariant distribution μ and transition probabilities P , the entropy rate is $\sum_{ij} \mu_i P_{ij} \log P_{ij}$.

See also Entropy is discussed in Chapter 2, and the Asymptotic Equipartition Property in Chapter 3 gives the most thorough characterization of entropy and compression. Chapter 4 is all about entropy rates of stochastic processes, and Chapter 5 is about data compression. Algorithms for constructing optimal codes for a probability model are discussed in Section 5 of these notes.

3 Predictive likelihood

Given a probabilistic model (that we may have fit given some training data), it is natural to score the model's ability to predict new data by evaluating the average likelihood the model assigns to held-out test data. To analyze that procedure in a probabilistic framework, we imagine that test data are generated from a true underlying distribution p , and we wish to evaluate the average predictive log likelihood that some fit model q assigns to that data.

Definition 3 (Predictive log likelihood). *The predictive likelihood of q on test data generated iid from p is*

$$\mathbb{E}_p[\log q(X)] = \sum_x p(x) \log q(x).$$

The same term already arose in Eq. (3) in the expression for average string length. In fact, the average codeword length is precisely related to the predictive log likelihood, as shown in the next claim.

Claim 3 (Predictive log likelihood is a model's ability to compress). *The average asymptotic codeword length when applying an optimal code designed for q to data generated by p satisfies*

$$\mathbb{E}_p[L_q(X)] = - \sum_{x \in \mathcal{X}} p(x) \log q(x)$$

Proof. Starting from Eq. (2),

$$\begin{aligned} \mathbb{E}_p[L_q(X)] &= H(p) + D(p||q) \\ &= - \sum_i p_i \log p_i + \sum_i p_i \log p_i - \sum_i p_i \log q_i \\ &= - \sum_{x \in \mathcal{X}} p(x) \log q(x). \end{aligned}$$

See C&T Thm 5.4.3 for a similar proof. □

For stationary stochastic processes, we instead evaluate $\lim_{n \rightarrow \infty} \mathbb{E}_p \log q(X_n | X_{n-1}, \dots, X_1)$.

See also MacKay (2003)[Chapter 28].

4 Entropy of continuous-valued processes

Entropy for continuous-valued random variables and processes is easy to define but it loses some of the nice interpretation with regards to compressibility.

Definition 4 (Differential entropy, C&T Section 8.1). *For a random variable X taking values in $\mathcal{X}$ with probability density function f , the differential entropy of X is*

$$h(X) := - \int_{x \in \mathcal{X}} f(x) \log f(x) dx$$

Example 1 (Uniform distribution, C&T 8.1.1). *If X is uniform on $[0, a]$ then its density is $1/a$ from 0 to a and 0 elsewhere, so*

$$h(X) = - \int_0^a \frac{1}{a} \log \frac{1}{a} dx = \log a.$$

If $a < 1$, then $h(X) < 0$.

Since differential entropy can be negative, it can't immediately be interpreted as a number of bits per sample (and to encode a continuous-valued sample it would take an infinite number of bits anyway).

Claim 4 (Interpretation of differential entropy, C&T Thm 8.3.1). *An n -bit quantization of X has entropy (and optimal average codeword length) $h(X) + n$.*

However, the definition of an n -bit quantization changes if we change the units in which X is measured; that is, differential entropy is invariant only to rigid transformations. For distributions over “features” which have no natural units, that destroys the interpretability of differential entropy.

Fortunately, relative entropy $D(f||g)$ maintains all of its good properties from the discrete case: it is always non-negative and it is invariant to all (measurable) transformations, including changing units and scales. In the context of predictive log likelihoods (as in Section 3), that corresponds exactly to comparing *differences* of predictive log likelihoods.

5 Constructing codes

Todo: model-based entropy coding with Huffman and arithmetic codes, model-free coding with Lempel-Ziv.

References

- Cover, Thomas M and Joy A Thomas (2006). *Elements of Information Theory*, 2nd edition. John Wiley & Sons. URL: <https://dl.dropboxusercontent.com/u/177449/books/Elements%20of%20Information%20Theory%20Cover%20and%20Thomas.pdf>.
- MacKay, David JC (2003). *Information theory, inference and learning algorithms*. Cambridge university press. URL: <http://www.inference.phy.cam.ac.uk/itprnn/book.pdf>.